

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

MAY/JUNE EXAMINATION

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS

WEDNESDAY, 21 MAY 2025

TIME: 2.00 PM – 4.00 PM (2 HOURS)

DIPLOMA IN SOFTWARE ENGINEERING

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 1**

- a) A data structure is an arrangement of data in a computer's memory or even disk storage. Provide **THREE (3)** data structure operations that are crucial for solving complex computer science problems. (9 marks)
- b) Serving as the foundation of a programming tool, a data structure illustrates the connections between data elements and enables programmers to efficiently manage the data. Identify **TWO (2)** categories of non-primitive data structures, explain each category, and provide **ONE (1)** example for each. (6 marks)
- c) Pseudocode is used to understand programming ideas and plans before writing actual code. Examine **TWO (2)** components of pseudocode and provide an example for each in your answer. (6 marks)
- d) An algorithm is a finite set of instructions designed to accomplish a particular task. Identify **TWO (2)** characteristics of an algorithm. For each characteristic, determine its role in ensuring the algorithm's efficiency in real-world applications. (4 marks)

[Total: 25 marks]

Question 2

- a) The Iterator interface provides a uniform way of traversing elements in various collections. Differentiate between the operations supported by *hasNext()* and *next()* in the Iterator. (4 marks)
- b) Indicate **THREE (3)** examples of counting primitive operations used to assess the running time of an algorithm without conducting experimental analysis. (3 marks)
- c) Representing algorithms through expressions allows for a clear understanding of their functionality. Apply **THREE (3)** types of analysis used in algorithms and demonstrate how each is used in algorithm performance evaluation. (9 marks)
- d) The efficiency of an algorithm is gauged by its order of growth. Provide **THREE (3)** notations used to classify the growth of an algorithm. (9 marks)

[Total: 25 marks]

Question 3

a)

```
1 import java.util.*;
2
3 public class SetExample {
4     public static void main(String[] args) {
5         Set<String> cities = new HashSet<>();
6         cities.add("Kota Kinabalu");
7         cities.add("Kuching");
8         cities.add("Ipoh");
9         cities.add("Kota Kinabalu");
10
11         for (String city : cities) {
12             System.out.println(city);
13         }
14     }
15 }
```

Figure 1: Code snippet using HashSet

- (i) Based on **Figure 1**, examine the codes to determine if any cities are added to the set more than once. (2 marks)
- (ii) Predict whether any city names will appear multiple times in the output. Include an explanation in your answer. (3 marks)
- (iii) Briefly explain the mechanism that a **HashSet** uses to ensure each element is unique. (2 marks)
- c) Compare and contrast **HashSet** and **LinkedHashSet** in terms of their ordering elements. (4 marks)
- d) A linked list is a dynamic data structure that offers programmers with the flexibility to adjust the memory consumption of software programs. Provide **THREE (3)** disadvantages of linked lists. (6 marks)
- e) Examine the **FOUR (4)** fundamental operations associated with the stack data structure and explain their purpose and functionality. (8 marks)

[Total: 25 marks]

AMCS2034 INTRODUCTION TO DATA STRUCTURES AND ALGORITHMS**Question 4**

- a) A graph is a mathematical structure that represents relationships among entities. Provide **THREE (3)** applications of graphs in the real world. (6 marks)
- b) Compare and contrast directed and undirected graphs, focusing on the directionality of edges. (4 marks)
- c) Graph traversal is the process of visiting each vertex in the graph exactly once. Identify the **TWO (2)** popular methods used to traverse a graph. (6 marks)
- d) Based on your answer in **Question 4 c)**, state **TWO (2)** common applications for each graph traversal method. (4 marks)
- e) Garbage collection is an important aspect of Java memory management. Classify **TWO (2)** advantages of garbage collection and identify **ONE (1)** type of garbage collection. (5 marks)

[Total: 25 marks]